

GlucoScreen-C: Smartphone Test Strips for Health Screening User Experience Demo

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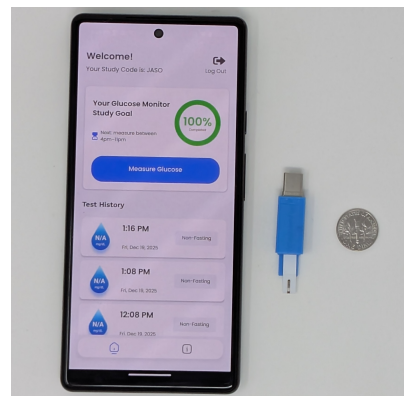
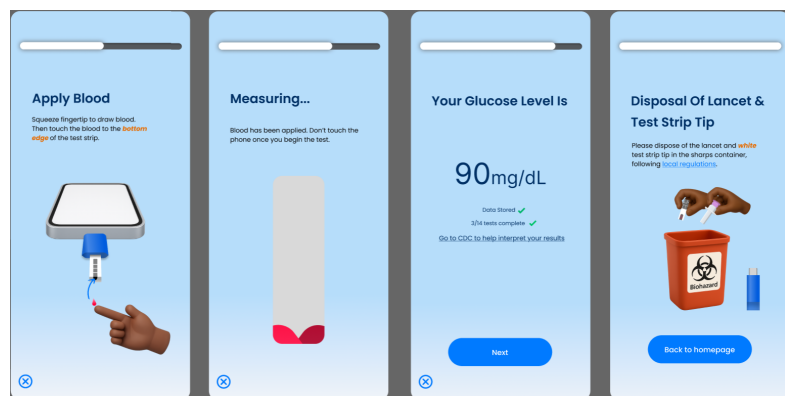


Figure 1: The GlucoScreen-C system includes a custom test strip that works directly with the smartphone and accompanying custom app that walks the user through a test. The demo will involve users experiencing a test for their first time.

Abstract

GlucoScreen-C, as a novel platform for smartphone-based test strips for biomarker screening, moves minimal hardware from the reader into the test strip device, allowing it to communicate the result of electrochemical biomarker detection reactions to the phone without incurring high cost or requiring extra hardware (beyond the test strip itself). This is the first of its kind, and starts to enable a new vision for healthcare where the smartphone becomes the center of testing. With its common sensors, large touchscreen user interface,

and familiar interaction patterns, this can become a much more approachable form of Rapid Diagnostic Testing (RDT) in the future. We invite the CHI community to try our prototype demo and share their thoughts and ideas about the future of remote healthcare.

CCS Concepts

• **Applied computing** → **Consumer health**; • **Hardware** → *Biology-related information processing*; • **Human-centered computing** → *Smartphones*.

Keywords

prediabetes, healthcare diagnostics, population health, rapid diagnostic testing, low-power, smartphones, ubiquitous computing



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1 Introduction

Glucoscreen-C is a new test strip that builds on earlier work on smartphone-based test strips to leverage the growing ubiquity of USB-C (which the iPhone adopted in 2024) and the audio channel of input to the phone's microphone to develop the cheapest proven transmission of an electrochemical signal generated by a biomarker test strip directly to the phone. It is small and intuitive to use, with a USB end and a sample end, connected by minimal circuitry, as shown in Fig. 2

Typical hardware for communicating data over USB-C requires a microcontroller to communicate the protocol. However, lower cost methods can potentially enable one-time-use accessories for the smartphone that can help us gather more data about our health. In our prototype test strip, we leverage the hardware of USB-C without the associated circuitry that increases cost. The first biomarker to be validated with this system in this study is glucose, but other electrochemical biomarkers are theoretically compatible with the system and could be developed.

2 Related Work

Despite the advent of continuous glucose monitoring (CGMs), the most common self-monitoring of blood glucose remain glucometers, due to their high accuracy and relatively low cost. Test strips include an electrochemical reaction region, which operates microfluidically to pull a drop of blood into a chamber. Glucose is oxidized and the amperometric change caused by the released electrons is measured by the standalone meter (Fig 3). Developments in new reactants reduced accuracy dependence on blood O_2 concentration from first generation meters, which used glucose oxidase, and enabled second generation meters, which use dehydrogenase and a mediator, often ferricyanide, for increased accuracy [2, 4, 6].

Within glucometers, there is variation on the accuracy and quality, which affects the usability and the price. Contour Next One and the Dario offer the most portable designs and connection to the phones, but their reliance on external readers and PCBs make them more costly and less suited for testing on an infrequent basis. With a starting cost of \$25 and \$45, respectively, the meter is still expected to be considered a loss leader, with the expected cost difference made up by profits on test strips over time. This leaves room for a fully standalone test strip that plugs directly into the phone as a potential lower-priced option that does not currently exist. This would need to be produced with an extremely low cost of manufacturing in order to provide a product with a sustainable and long-lasting business model to provide that entry point for new or infrequent users [7].

Noninvasive methods of glucose detection have been proposed for decades. However, accuracy remains too low for commercial use and a viable principle of sensing is yet to be determined, though some progress with validation on small datasets has been shared

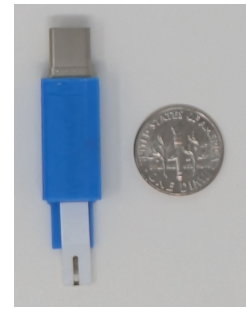


Figure 2: The GlucoScreen-C test strip prototype. One end plugs into the USB-C port of a modern smartphone and the other end accepts the user's sample.

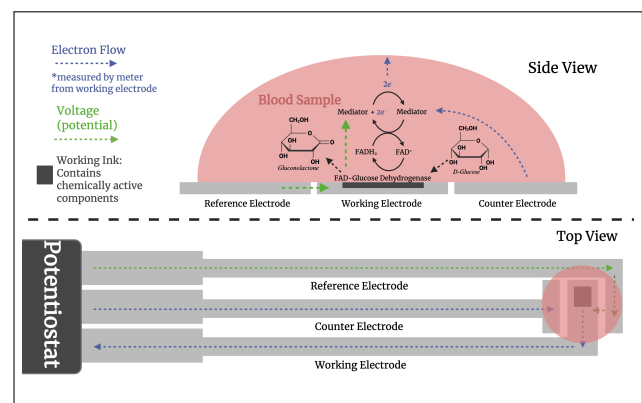


Figure 3: 2nd gen FAD-glucose dehydrogenase pathway

[1, 5]. If proven possible, this would represent a significant advance, but nothing has been commercially released yet with accuracy that can be used for medical screening.

3 Methods

3.1 Test Strip Hardware Design and Rationale

The test strip hardware has been designed towards the lowest-cost method to accurately transduce the signal from an electrochemical reaction to a smartphone via commodity sensors. USB-C was chosen as both the mechanical interface and power source because it is found on virtually all smartphones and provides stable power via USB default mode. For the prototype, commercially available test strips (True Metrix) were used and placed in contact with PCB traces. For volume production, it would be feasible to glue the biochemical reaction chamber directly onto the PCB traces using conductive adhesive, creating a one-time use test strip and further reducing the cost of production.

We have a new circuit design that it is a standalone test strip that translates an electrochemical reaction, which responds to glucose in a sample, to a format that a smartphone can read via an app and common smartphone sensor [7]. However, the signal for the new GlucoScreen-C test strip differs from prior work in its compact form and ease of use, enabled by its novel method of transmission. By

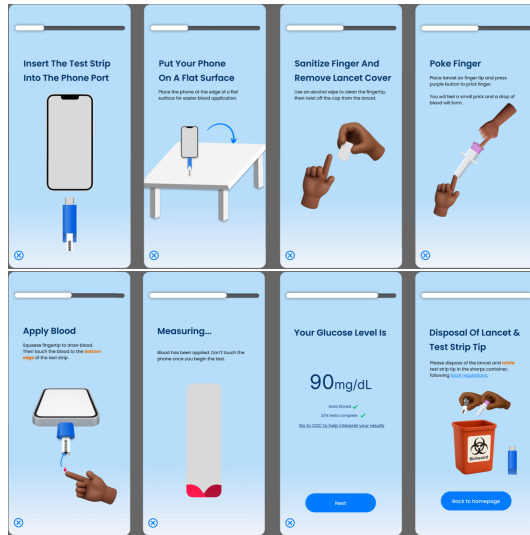


Figure 4: A visual animated guide walks the user through the process of using the test strip to collect a blood sample.

transforming the electrochemical signal from the glucose test strip reaction end into an audio signal, it can theoretically be designed to be more compact and lower cost. The signal is transduced via a potentiostat into a speaker driver that is filtered prior to emission. The sound is emitted at a high frequency which enables it to be outside the typical audible range for humans. This frequency is adjustable via interchanging resistor and capacitor values to fit regulations and comfort from user testing and further development.

3.2 Custom Smartphone App Architecture

GlucoScreen-C was developed to provide visual guidance and feedback for standardized blood glucose testing without requiring clinical supervision. The system combines instructional animations visualized task elements trend views testing history and progress indicators to support understanding and adherence across repeated use. The design process followed the finger prick sampling procedure including equipment preparation, fingertip cleaning, lancet use, blood expression, and strip contact. Each step was analyzed to identify actions that required explicit visual explanation and was translated into structured on screen guidance. Icons, illustrations, and hand action diagrams were primarily designed in Figma with additional assets generated using AI models and manually refined for consistency. Visuals emphasized high contrast simple line work and uniform styling to remain legible under low vision or constrained conditions. Key actions were animated frame by frame and exported using vector based animation tooling to ensure low performance overhead. Animation timing was calibrated to reflect real world actions and avoid misleading pacing. The interface presents glucose readings across multiple time ranges with visual distinctions between normal and elevated values. Early testing showed that certain labels increased anxiety for first time users. Terminology and symbols were adjusted to reduce cognitive load and improve clarity while preserving informational meaning. Multiple rounds of usability testing were conducted in which participants

relied solely on in app visual guidance to complete the sampling process. Observed issues such as uncertainty during lancet use or difficulty aligning blood drops informed iterative updates. Refinements included clearer transitions emphasis cues and preventative prompts designed to improve safety confidence and task success.

The GlucoScreen-C system follows a modular event driven architecture that separates mobile workflows from backend services. Client side interaction data ingestion processing and persistent storage are handled as distinct components to improve maintainability and allow independent updates as system requirements evolve.

The client application was implemented using React Native with an ejected Expo configuration to enable cross platform deployment while retaining access to native device capabilities such as audio sensors motion sensors and secure storage. Functional React patterns were used to separate interface presentation from application logic.

Front end interfaces were designed in Figma and implemented to support structured testing workflows and real time system feedback. Separate study and release interface variants were maintained to enable usability testing without exposing finalized behavior or outputs during evaluation.

Navigation and screen level state were managed through a declarative routing framework to ensure predictable user progression through the testing workflow. Simple key value storage was used to persist non sensitive information such as study identifiers and session flags across application launches.

Raw uploads are stored in Amazon S3 while processed outputs and associated metadata are stored in DynamoDB to support low latency retrieval of recent and historical test entries. Lambda functions serve as the API layer handling request validation and response formatting through API Gateway.

All backend communication uses encrypted channels and access to stored health related data is restricted through tightly scoped permissions consistent with HIPAA aligned practices. Error handling was implemented across client server interactions to detect failed uploads invalid inputs and backend execution errors and to surface appropriate retry guidance to users.

Backend services were implemented using a combination of AWS API Gateway, AWS Lambda, Amazon S3, and Amazon DynamoDB. User authentication requests are handled through API Gateway endpoints invoking Lambda functions, which validate credentials and establish authenticated sessions. User session and authentication metadata are stored in DynamoDB to support secure request handling.

Upon completion of a blood glucose test, data generated by the mobile application is uploaded to Amazon S3 using pre-signed URLs. This approach allows the client to upload test artifacts directly to storage without exposing write permissions, reducing attack surface while accommodating intermittent connectivity. Uploaded data initiates serverless processing pipelines implemented with AWS Lambda functions, which perform basic validation and transformation prior to downstream handling.

Collected data included demographics, survey responses, interview recordings, and biomarker measurements from both the prototype device and commercial meters. Prototype data were securely uploaded via the app, and commercial meter results were recorded using participant IDs. All data were anonymized or encrypted and

Concentration	FDA Range	Trials
46	1	3
118	2	3
333	3	3
480	>5	3

Table 1: Table of glucose concentrations tested in lab, with results displayed in Figure 5

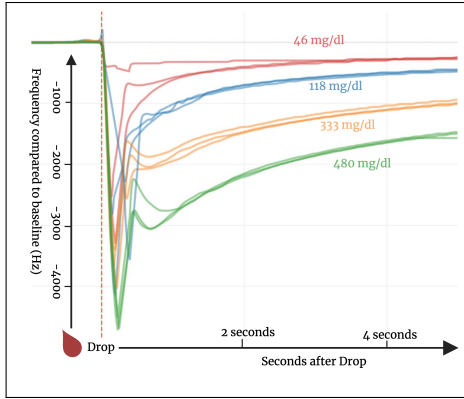


Figure 5: Results with control solutions in lab validate that this proof-of-concept is functionally working.

used solely for research purposes. The research team oversaw data integrity, storage, and access control.

3.3 Lab Testing Procedure for Feasibility and Separability

In lab, we tested for feasibility of our early hardware prototype to determine if this method is feasible. We used TrueMetrix control solutions, composed of water, d-glucose, buffers, viscosity enhancing agent, salts, dye and preservatives. Four different concentrations of control solutions were prepared via titration to the concentrations shown in Table 1.

First, the control solutions were tested using the TrueMetrix glucometer, and a dilution was made for the desired concentration. The directions on the app were followed, skipping the directions for taking a real blood sample. A drop from the diluted concentration was inserted into the 2nd generation test strip via a sterile transporter (aluminum foil). The test strip was then taken out and properly discarded. Many trials of each concentration were done with the same methods without reusing the same drops.

In order to verify feasibility of the hardware, software was written to extract the maximum frequency, establish the high carrier stable frequency and locate the point of the drop of blood by comparing successive points to a threshold. The maximum frequency was then plotted and compared at the baseline values for various concentrations. For the purposes of visualization, interpolation was used to predict the expected points where results could not be calculated.

4 Results

4.1 Accuracy Testing Results

A small amount of data was collected in lab to validate basic hardware and software pipeline functionality. As seen in Figure 5, we are able to visually distinguish the signals emitted by the test strip at different concentrations on repeated replicates. Further work on gathering more data and developing automated machine learning methods will allow accuracy validation against standards, such as the FDA guidelines [3].

4.2 The Demo

We have a few working prototypes of our test strip device, and we would like to share them with the community to try out the next generation test strips. We believe the hardware, combined with software animations in close proximity to the action, can improve the experience of using the test strips to gather important information about users' health.

We will have a table set up with phones and strips, as well as a QR code to access the app on your own phone to try it out. In addition, we'll be on hand to discuss the demo and chat with attendees about how the device could be thought about in the context of their own health journey.

5 Discussion

This demonstration of prototype technology for healthcare represents the first step into universal smartphone-based test strips. We hope that a demo and subsequent user reflection will help us hone the design and direction of the project and enable this project and authors to empower users to collect data about their own health one day.

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